SpectralZero: Text-Driven Spectral–Spatial Alignment for Zero-Shot Hyperspectral Image Classification

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***Abstract*—Recently, deep learning-based hyperspectral image (HSI) classification methods have witnessed significant advancements. However, existing approaches predominantly focus on feature modeling for seen classes during the training phase, while the exploration of generalization and transfer mechanisms toward unseen land-cover categories remains limited. As a result, these methods exhibit poor adaptability and generalization performance in zero-shot scenarios. To address this challenge, we propose a novel zero-shot HSI classification framework called SpectralZero that leverages the strong generalization capability of large language models (LLMs) to compensate for the absence of semantic information in unseen classes. Moreover, our framework explicitly captures the synergistic relationship between spatial and spectral features to facilitate knowledge transfer from seen to unseen categories. Specifically, the proposed framework comprises two key modules: semantic prompt enrichment (SPE) and spectral–spatial union extraction (SSUE). SPE utilizes LLMs to generate fine-grained semantic descriptions of land-cover classes, mitigating the semantic representation deficiency caused by the lack of samples in unseen categories. SSUE introduces two separate branches to independently model spatial structures and spectral reflectance characteristics, thereby enhancing the model's capacity to extract joint spatial–spectral representations. A major advantage of our method lies in its ability to accurately classify unseen classes without requiring any samples during training or inference. Experiments on multiple public HSI datasets demonstrate that our approach consistently outperforms state-of-the-art methods in zero-shot classification tasks, exhibiting superior recognition accuracy and generalization performance. The code will be available at https://github.com/xiayang124/SpectralZero**

***Index Terms*—Cross-modal alignment, hyperspectral image (HSI) classification, zero-shot classification.**

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NOMENCLATURE

D	HSI dataset D with image I and label set L.
I	HSI $I \in \mathbb{R}^{H\times W\times C}$.
H	Height of the HSI.
W	Width of the HSI.
C	Number of spectral bands.
L^N	Set of all class labels, $L^N = \{l^1, l^2, \ldots, l^N\}$.
p	Patch size; each patch is of size $p \times p$.
I^P	Image patch cropped from I, $I^P \in \mathbb{R}^{p\times p\times C}$.
S	Seen classes.
U	Unseen classes.
I_S^P	HSI patch from the seen classes subset.
I_U^P	HSI patch from the unseen classes subset.
D^P	Patch-level dataset constructed from D, $D^P = \{I^P, L\}$.
D_S^P	Seen classes subset, $D_S^P = \{I_S^P, L_S^K\}$.
D_U^P	Unseen classes subset, $D_U^P = \{I_U^P, L_U^M\}$.
K	Number of seen classes.
M	Number of unseen classes.
L_S^K	Label set of seen classes, $L_S^K = \{l_S^1, l_S^2, \ldots, l_S^K\}$.
L_U^M	Label set of unseen classes, $L_U^M = \{l_U^1, l_U^2, \ldots, l_U^M\}$.
f_t	Pretrained CLIP text encoder.
o^t	Textual semantic feature extracted by f_t.
o^{spe}	Spectral feature extracted by the spectral branch.
o^{spa}	Spatial feature extracted by the spatial branch in SSUE.
S^{spe}	Spectral cross-modal embedding space.
S^{spa}	Spatial cross-modal embedding space.
M^{spe}	Similarity matrix in the spectral embedding space.
M^{spa}	Similarity matrix in the spatial embedding space.

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M	Unified similarity matrix after weighted fusion of M^{spe} and M^{spa}.
M_j	jth similarity score in the denominator of (3).
o^i	Spectral–spatial joint feature.
p^{cls}	Confidence (probability) vector for seen class prediction.
o^{spe_p}	Projected spectral feature in the spectral–text space.
o^{spa_p}	Projected spatial feature in the spatial–text space.
o^{t_p}	Projected text feature in the alignment space.
W_{spe}	Spectral projection matrix.
W_{spa}	Spatial projection matrix.
W_t	Text projection matrix.

I. Introduction

HYPERSPECTRAL images (HSI) offer rich spectral and spatial information, providing superior discrimination of land-cover types compared to traditional multispectral or RGB imagery [1]. Their continuous high-resolution spectral bands enable effective differentiation of visually similar classes, such as synthetic and natural grass, improving classification accuracy and reliability. HSI classification facilitates joint spectral–spatial feature representation, supporting applications like water body classification [2], plant monitoring [3], and mineral identification [4]. However, the high dimensionality of HSI poses significant challenges for feature extraction and classification, making efficient utilization of such data a key research focus.

Early HSI classification methods primarily relied on manually designed spectral–spatial feature extraction techniques, employing classical machine learning algorithms such as support vector machines (SVMs) [5], random forests (RFs) [6], and K-nearest neighbors (K-NNs) [7]. However, these approaches, constrained by expert heuristics, often fail to yield optimal representations. With the advent of deep learning, neural networks have shown a remarkable capacity for automated feature learning, significantly advancing classification performance. Chen et al. [8] introduced a 3-D convolutional neural network (3D-CNN) to jointly exploit spectral and spatial information, achieving superior results. Zhong et al. [9] enhanced model stability and expressiveness through deeper architectures and residual connections. More recently, Hong et al. [10] adopted the transformer's global attention mechanism to model long-range spectral dependencies, effectively addressing the locality limitations of conventional convolutional models and further boosting classification accuracy.

Despite the significant progress achieved in HSI classification, existing models still heavily rely on large amounts of labeled data for training and evaluation. In practical applications, a single HSI scene often covers extensive geographic areas and contains diverse and complex land-cover types. The prohibitively high cost of manual annotation renders comprehensive labeling infeasible, resulting in severe scarcity or even complete absence of labeled samples for certain classes [11]. Moreover, surface variations induced by geological activities or biological evolution, though effectively captured by sensors, are difficult to annotate continuously due to the high labor demands of long-term manual labeling, thereby exacerbating the emergence of zero-shot scenarios. In such cases, traditional classification models often struggle to accurately model the feature representations of unseen classes, leading to their misclassification as known categories and significantly impairing overall classification performance and generalization. Consequently, developing HSI classification methods that are robust under limited-sample or even zero-shot conditions has become an urgent and critical research challenge.

To alleviate the issue of high annotation costs, researchers have proposed several limited-sample HSI classification methods. The goal of limited-sample classification is to achieve considerable classification performance but relying solely on very limited labeled HSI samples (e.g., as few as ten samples per class for training). Liu et al. [12] constructed a 3D-CNN to extract HSI features and learn a metric space, and combined it with a nearest-neighbor classifier at the testing stage to perform HSI classification, achieving significant performance improvement in limited-sample scenarios. To mitigate the risk of model overfitting under limited data conditions, Xue et al. [13] proposed a more lightweight network architecture and introduced positive and negative samples across different scales to enhance the generalization capability of the model. Moreover, considering that natural images (such as ImageNet [14]) possess substantially richer labeled data compared to HSI, Li et al. [15] first pretrained a general spatial feature extraction network using natural images, subsequently training a dedicated spectral classification branch with limited HSI samples. Nevertheless, these methods remain inadequate for effectively addressing situations in which certain categories are entirely absent from the training labels.

In recent years, zero-shot classification—originally introduced in [16] to recognize unseen categories via language-based attributes—has garnered increasing attention. In this paradigm, the label spaces of training and testing are mutually exclusive, where the seen classes represent the categories available during training and the unseen classes denote novel categories that have never been observed by the model but can be inferred through learned semantic relationships. Mainstream approaches leverage image–text datasets such as ImageNet [14], CUB [17], and AwA [18], projecting visual and textual features into a shared semantic space to enable cross-modal alignment. Representative methods include semantic output codes [19], which map images into semantic embeddings for nearest-neighbor classification, and deep learning-based approaches [20], [21] that exploit large-scale data to learn complex semantic relations. Generative models such as GAN-based frameworks [22] further improve generalization by synthesizing features for unseen classes. However, these models often rely on limited training data and small-scale architectures, constraining their semantic modeling capacity. With advances in GPU computing and the rise of large-scale pretrained models, traditional datasets are increasingly

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inadequate. To address this, CLIP [23] was proposed, leveraging large-scale web-sourced image–text pairs and contrastive learning to construct a unified semantic space, which has since been extended to tasks like semantic segmentation [24], image generation [25], [26], point cloud classification [27], and anomaly detection [28]. Nonetheless, these approaches are primarily developed for RGB imagery and lack the capacity to capture rich spectral features in HSI, limiting their generalization in such domains.

Several early efforts have explored zero-shot classification in HSI. Yue et al. [31] proposed a spectral–spatial reconstruction method to enhance discrimination between seen and unseen classes. However, this approach treats all unseen categories as a single group, lacking fine-grained differentiation. Pang et al. [32] introduced a CLIP-based pseudo-labeling framework that mitigates noise through augmentation and refinement, improving stability without requiring manual annotation. Yet its exploitation of textual semantics remains limited, constraining visual-semantic representation learning. To further exploit the semantic transfer capability inherent in textual information, Zhang et al. [30] proposed using textual descriptions as auxiliary information to facilitate classification capability transfer between different datasets. Although this approach effectively leverages side information, it still relies on class labels during training and is restricted to scenarios where classes across datasets are identical, making it challenging to extend to zero-shot classification settings. To overcome dataset-specific training dependencies, Li et al. [29] developed HyperFree, which utilizes a spectral-wide dictionary and minimal prompts to generalize across HSI datasets. Despite its effectiveness, HyperFree still depends on class-specific prompts at inference, thus failing to identify unseen categories in strict zero-shot settings. A comparison of these approaches has been shown in Fig. 1. In summary, the current research on zero-shot classification of HSI remains at an exploratory stage and faces several notable limitations.

1) Most mainstream ZSC methods are primarily trained on RGB imagery, failing to fully model and exploit the rich spectral information inherent to HSIs, thereby limiting their generalization capability in hyperspectral scenarios.
2) Existing HSI-ZSC models (see [32]) exhibit inadequate integration and utilization of auxiliary information—such as textual semantics or class attribute descriptions—thus, underutilizing the potential of semantic knowledge in enhancing class discrimination and knowledge transfer.

To tackle the aforementioned challenges, this article proposes a novel framework for HSI zero-shot classification, termed semantic-driven spectral–spatial alignment (SpectralZero). SpectralZero enables effective recognition of unseen land-cover categories by leveraging labeled samples from seen classes and associated auxiliary semantics, addressing limitations due to data scarcity or annotation costs. The framework comprises two key components: the semantic prompt enrichment (SPE) module and the spectral–spatial union extraction (SSUE) module. SPE module enriches category representations with visual attribute descriptions to enhance semantic generalization. SSUE module introduces a

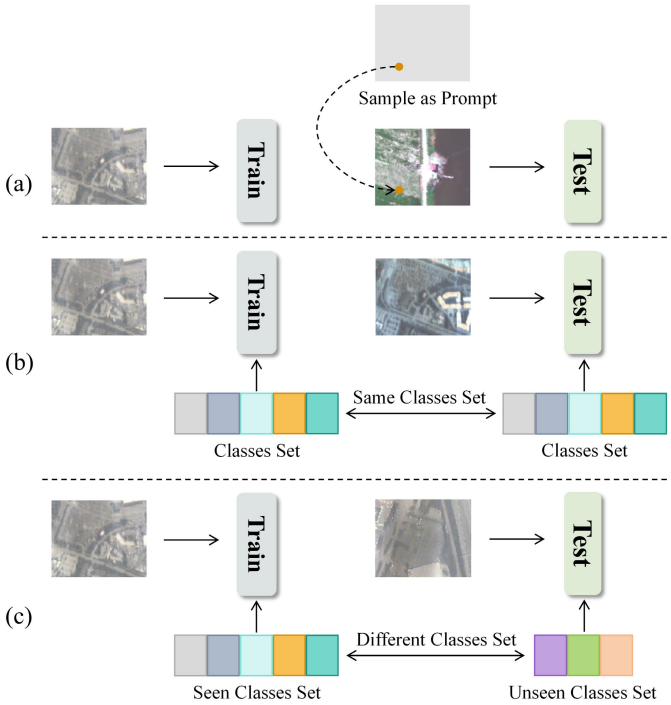


Fig. 1. Difference among three models. (a) Prompt learning-based zero-shot classification (such as HyperFree [29]), which still requires class-related samples as prompts for testing; (b) domain generalization (see [30]), which is conducted within the same dataset and category settings; and (c) text-guidance zero-shot classification (ours), where the train and test classes are disjoint, and the model identifies different land-cover categories through cross-class knowledge transfer.

decoupled spectral feature extraction pathway, where global spatial pooling suppresses structural interference, and dual projection spaces compute image–text similarity matrices from spatial and spectral features independently. These are fused via proportional weighting to form a unified similarity matrix for contrastive learning, thereby improving cross-modal alignment and unseen class classification. Together, the SPE module and SSUE module synergistically strengthen the model's capacity for semantic modeling and spectral representation, facilitating robust generalization to novel categories.

1) This article proposes a novel HSI classification framework called SpectralZero tailored for zero-shot scenarios. By leveraging the generalization capability of large language models (LLMs), the framework effectively addresses the information deficiency of unseen classes. Furthermore, it exploits spatial–spectral features to facilitate knowledge transfer from seen to unseen categories. Experimental results on multiple publicly available datasets demonstrate that the proposed framework outperforms existing zero-shot classification methods, validating its effectiveness and generalizability.
2) An SPE module is introduced to generate fine-grained semantic descriptions of land-cover classes based on LLMs, mitigating the lack of visual information for unseen categories. In addition, the module embeds auxiliary semantic information into a unified semantic space via the CLIP model, thereby constructing implicit semantic associations between seen and unseen classes.

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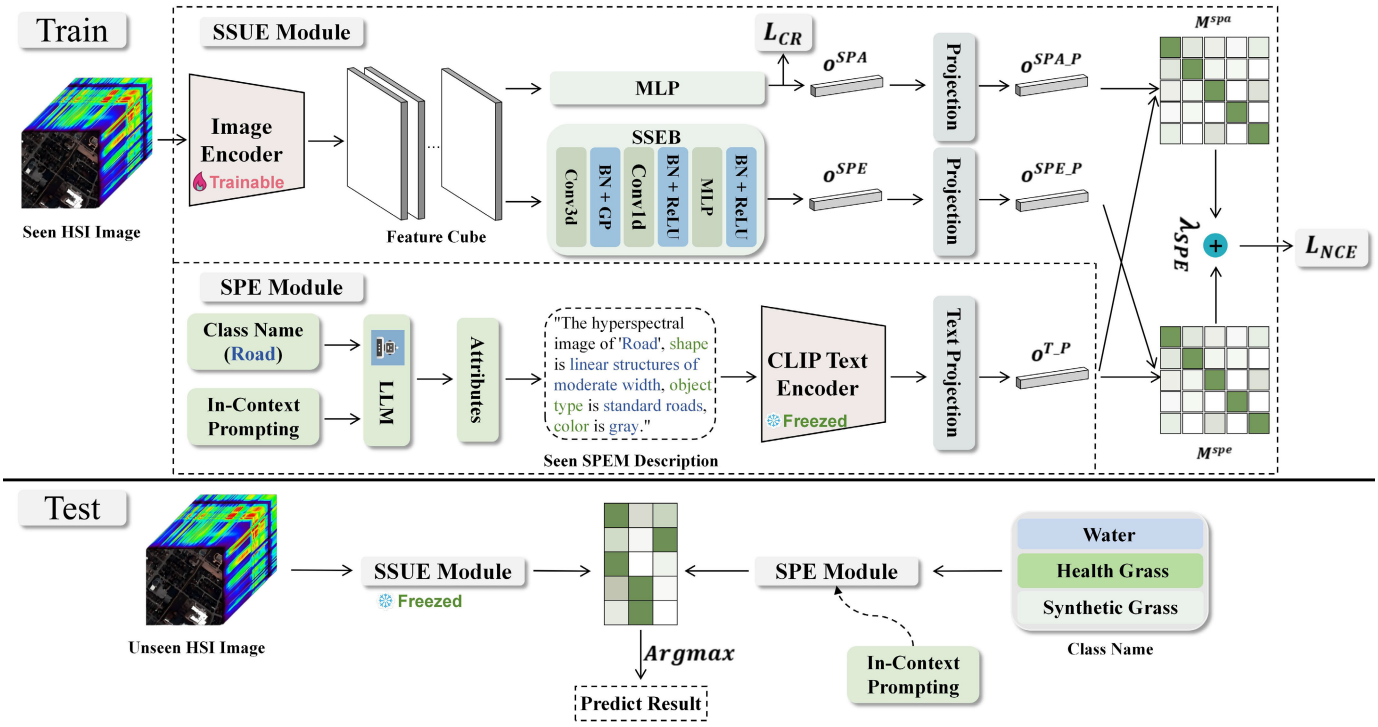


Fig. 2. Framework of the proposed SpectralZero. The SPE module encodes rich natural language descriptions into semantic features and projects them into both spatial and spectral embedding spaces. The SSUE module decouples image features into spatial and spectral components, which are then aligned with the corresponding semantic features. The similarity matrices from the two aligned spaces are fused and used to compute the loss during training. During testing, classification is performed by selecting the class index corresponding to the maximum value in the fused similarity matrix.

This significantly enhances the model's capacity for semantic modeling and recognition, providing critical support for zero-shot classification tasks.

3) An SSUE module is designed to improve the model's ability to extract spatial and spectral features under zero-shot conditions. The module consists of two independent branches: a spatial branch for capturing structural information of land covers, and a spectral branch for explicitly modeling the rich reflectance characteristics inherent in hyperspectral data. These features are then deeply fused to generate a unified embedding capable of jointly expressing spatial and spectral attributes, such as color and material properties, thereby significantly enhancing classification performance in zero-shot scenarios.

II. Methodology

This section presents the overall architecture of the proposed SpectralZero framework and provides a detailed description of its two core components: SPE module and SSUE module. In addition, Section II-D further elaborates on the implementation details of the model. The overall pipeline of SpectralZero is illustrated in Fig. 2.

A. Overview

Given an HSI dataset $D = \{\mathbf{I}, \mathbf{L}\}$, where $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$ denotes an HSI, H and W represent the height and width of the image, and C is the number of spectral bands. The set of labels is denoted as $L^N = \{l^1, l^2, \ldots, l^N\}$. To facilitate model training and evaluation, we extract image patches of size $p \times p$ centered on specific pixels from image $\mathbf{I}$ in dataset D, and associate them with corresponding class labels to construct a subset $D^P = \{\mathbf{I}^P, \mathbf{L}\}$, where $\mathbf{I}^P \in \mathbb{R}^{p\times p\times C}$. According to the zero-shot classification setting, the subset D^P is further divided into seen classes subset $D_S^P = \{\mathbf{I}_S^P, \mathbf{L}_S^K\}$ and unseen classes subset $D_U^P = \{\mathbf{I}_U^P, \mathbf{L}_U^M\}$, where S and U denote the seen and unseen classes, K and M denote the number of seen and unseen classes, respectively. Each sample from seen classes subset contains an image patch $\mathbf{I}_S^P \in \mathbb{R}^{p\times p\times C}$, a label set $\mathbf{L}_S^K = \{l_S^1, l_S^2, \ldots, l_S^K\} \in \mathbf{L}^N$. The construction of the unseen class subset is similar to that of the seen classes, including image patches $\mathbf{I}_U^P \in \mathbb{R}^{p\times p\times C}$, the unseen label set $\mathbf{L}_U^M = \{l_U^1, l_U^2, \ldots, l_U^M\} \in \mathbf{L}^N$.

This article focuses on the strict zero-shot classification setting, where the label sets of seen and unseen classes are disjoint and satisfying

$$\mathbf{L}_S^K \cap \mathbf{L}_U^M = \emptyset. \tag{1}$$

The goal of strict zero-shot classification is to ensure that no unseen class samples participate in training. Under this constraint, only the seen class subset D_S^P is used as training data to learn a mapping function

$$f : \mathbf{I}_S^P \rightarrow \mathbf{I}_U^P \tag{2}$$

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TABLE I
HYPERPARAMETER DEFINITIONS

Symbol	Description
τ	Learnable temperature parameter in InfoNCE loss.
B	Batch size used in training.
λ_{sim}	Spectral similarity fusion coefficient, $\lambda_{\text{sim}} \in [0, 1]$.
λ_{loss}	Loss fusion coefficient, $\lambda_{\text{loss}} \in [0, 1]$.

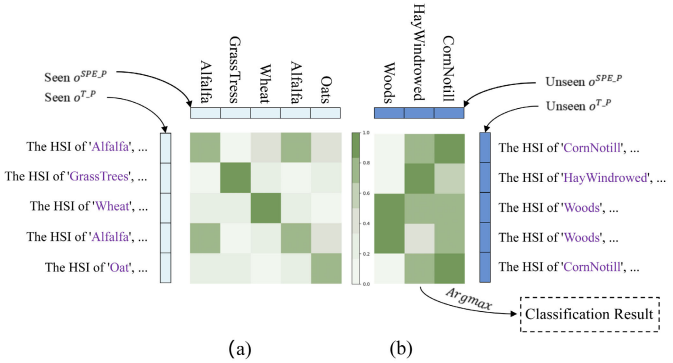


Fig. 3. Similarity matrix of the proposed model in both the training and testing stages. (a) Training stage. The features from the two different modalities are aligned using cosine similarity to generate a similarity matrix. Higher similarity corresponds to a darker color in the matrix. (b) Testing stage. The similarity matrix is generated in the same manner as in the training stage, and prediction results are obtained by selecting the indices corresponding to the maximum values.

such that the function f enables accurate classification on the unseen subset D_U^P. For clarity, we summarize key notations and hyperparameters in Nomenclature and Table I.

First, the proposed SpectralZero framework consists of two main stages: training and testing. During the training stage, the class name and corresponding HSI data $\mathbf{I}_S^P$ from the seen classes are input into the *SPE module* and the *SSUE module*, respectively. Specifically, the SPE module leverages the rich knowledge and contextual reasoning capability of pretrained LLMs, guided by in-context prompts, to automatically generate semantically enhanced textual descriptions that provide richer visual-semantic expressions than the original class names. Subsequently, these semantic-enhanced texts are fed into the pretrained CLIP text encoder f_t to accurately extract textual semantic features o^t with stronger discriminative capability. Meanwhile, the SSUE module initially extracts features from the HSI patch, and the resulting low-dimensional features are separated into spectral and spatial branches. Each branch independently performs 1-D spectral feature extraction and spatial feature extraction with auxiliary classification, outputting spectral features o^{spe} and spatial features o^{spa}, respectively. Subsequently, the extracted spatial features o^{spa} and spectral features o^{spe} are jointly input into the image projection to align with textual semantic features o^t, thereby constructing two cross-modal embedding spaces, S^{spa} and S^{spe}. These visual projections are matched with text semantic projections o^t to generate two similarity matrices, M^{spa} and M^{spe}, as depicted in Fig. 3(a). Finally, a weighted fusion strategy is applied to combine these two similarity matrices into a unified similarity matrix M. This unified similarity matrix is then optimized via the InfoNCE loss function [33]. The formula of the InfoNCE loss is defined as follows:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(M/\tau)}{\sum_{j=1}^{B} \exp\left(M_j/\tau\right)} \tag{3}$$

where τ is a learnable temperature parameter that controls the smoothness of the contrast during training and B is the batch size.

During the testing phase, the semantic-enhanced texts and HSI data from the unseen class dataset are, respectively, fed into the SPE module and the SSUE module to obtain the corresponding text features and dual-branch spectral–spatial features. These three features are then aligned to generate a fused similarity matrix, as illustrated in Fig. 3(b). The model determines the predicted class of each sample by selecting the index of the maximum value in each row of the fused similarity matrix, thereby completing the zero-shot classification task for HSI.

B. SPE Module

This section introduces the SPE module, which expands class labels into attribute-level natural language descriptions with scale-consistent cues to enhance textual expressiveness for cross-modal alignment. Specifically, taking the IP dataset, we first define multiple visual attribute dimensions for every class, including shape (e.g., land surface), object type (e.g., crop type), and color (represent spectral information, e.g., vegetation color), and provide relevant visual descriptions for each attribute. Fine-grained class descriptions are generated by a pretrained LLM using a fixed prompt template and no post-processing (e.g., cleaning or filtering) and no manual curation are applied. For each dataset, we use the same prompt template to ensure consistent semantic generation across classes and datasets. Then, using a class-prepending strategy, the class name is combined with each visual attribute description which are listed in Table II to generate a comprehensive SPE text $\mathbf{T}$ that systematically conveys attribute-level class semantics, the prompting template and the attribute processing flow are shown in Fig. 4.

To enhance the expressive ability of semantic prompts and ensure the accuracy and richness of the generated texts in capturing visual attributes of land-cover categories, SPE further utilizes a pretrained LLM to automatically generate fine-grained attribute descriptions covering multiple visual dimensions, such as color, shape, material, and structure, based on class names and in-context prompts. Importantly, the text generation for seen and unseen classes uses only the class names and dataset-agnostic domain knowledge, and does not access any pixels, masks, statistics, geographic metadata, or image-derived cues from the target datasets. Thus, no visual or dataset-specific information from unseen classes is leaked into training. This strategy not only significantly expands the semantic representation capacity of textual prompts but also establishes more precise and discriminative distinctions among classes within the semantic embedding space. By leveraging this semantic enrichment mechanism, the generated textual descriptions achieve higher quality alignment with visual features in the semantic space, thereby effectively enhancing the

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TABLE II
VISUAL ATTRIBUTE DESCRIPTIONS (IP DATASET FOR EXAMPLE)

Class	Attributes	Description	Text
CornNotill	**crop type** land surface other attributes	**corn** flat green	The hyperspectral image of 'CornNotill', crop type is corn, land surface is flat, vegetation color is green.
SoybeanMintill	crop type **land surface** other attributes	soybean **undulating** mintill	The hyperspectral image of 'SoybeanMintill', crop type is soybean, land surface is undulating, vegetation color is green.
CornMintill	**crop type** **land surface** other attributes	**corn** **undulating** mintill	The hyperspectral image of 'CornMintill', crop type is corn, land surface is undulating, tillage is mintill.

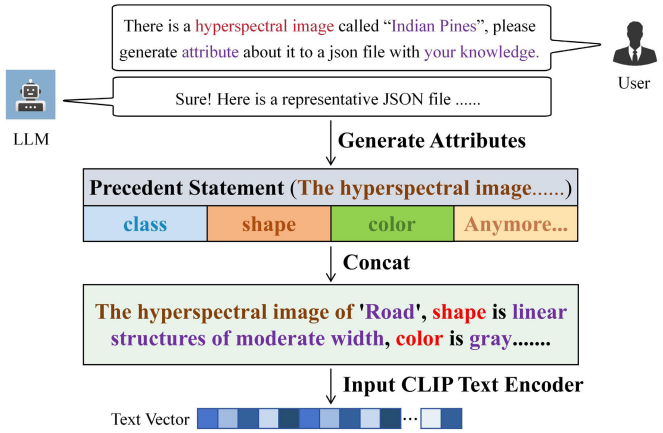


Fig. 4. Semantic enhancement text generation workflow. We first provide the LLM with a prompt containing key terms and ask it to generate attribute-level descriptions, such as shape, color, and vegetation category. These attributes are then concatenated with a predefined template to form an enriched semantic prompt, which is subsequently fed into the CLIP text encoder to obtain reliable textual feature representations.

model's understanding of image content and its discriminative performance in zero-shot scenarios. Fig. 5 further illustrates the process by which the model transfers knowledge from seen to unseen classes through linguistic information.

To precisely capture the visual-semantic information embedded in the SPE text, we further introduce the CLIP text encoder which is frozen to encode the enriched descriptions. The CLIP model, trained on large-scale image–text pairs, possesses strong capabilities in textual understanding and semantic representation. By leveraging the pretrained CLIP text encoder, the model can extract accurate and discriminative textual semantic features. Subsequently, the extracted textual features are projected by a mapping head into two separate cross-modal embedding spaces within SSUE. These embeddings are then aligned with the spatial and spectral features of the image, respectively, to enable feature alignment and cross-modal classification.

C. Spectral–Spatial Union Feature Extraction Module

Although the preliminary 3D-CNN demonstrates strong capability in learning spectral–spatial joint representations, its output features are inherently entangled in terms of spatial information and spectral responses. When such entangled representations are directly projected into a unified cross-modal embedding space, variations in spatial context may interfere with spectral semantics, resulting in semantic misalignment and reduced discriminability of cross-modal alignment. Meanwhile, this coupling makes it difficult to explicitly and controllably adjust the relative contributions of spatial and spectral information during alignment. Therefore, we propose the SSUE module to decouple the initial features and separately extract spatial information and spectral response characteristics, providing a more controllable representational basis for subsequent alignment and fusion. The proposed module comprises four subcomponents: a preliminary image encoder, a 1-D spectral extraction branch, a spatial extraction and classification branch, and a dual-branch alignment similarity-weighted fusion mechanism.

The preliminary image encoder is adopted from [30], comprising multiple 3-D convolutional blocks designed to extract spectral–spatial joint features o^i from input images. However, this representation inherently mixes spatial and spectral information, adversely affecting the precise extraction of image spectral features. Thus, we introduce a single-dimensional spectral extraction branch (SSEB) within the SSUE module, aiming to reduce interference from spatial image features and enhance the spectral extraction capability of the model. Specifically, this spectral extraction branch initially performs dimensionality reduction on preliminary features o^i (extracted by the preliminary image encoder) through a 3D-CNN layer, generating intermediate features with dimensions $C' \times H' \times W'$. Subsequently, global pooling is applied to aggregate these intermediate features along spatial dimensions into a 1-D spectral feature vector o^{spe}. Next, the spectral vector undergoes feature refinement through 1D-CNN layers to extract local spectral features, followed by additional BN layers for capturing nonlinear spectral characteristics. Finally, the resultant spectral features o^{spe} are projected into the spatial–text cross-modal union embedding space S^{spe} for subsequent cross-modal alignment and similarity-weighted fusion.

Concurrently, to effectively extract and classify spatial features, the spectral–spatial joint features o^i obtained by the

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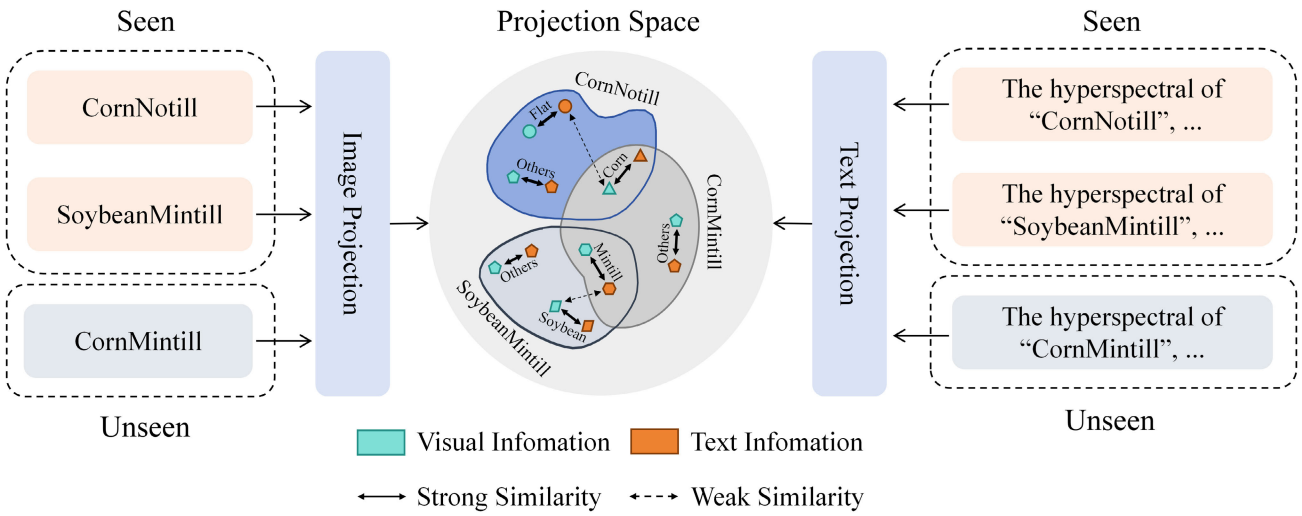


Fig. 5. Visualization of the projection space where visual and text features are aligned. Visual features and textual features are mapped into a shared projection space. Within this space, green shapes represent visual information and orange shapes denote textual information, with different geometric shapes used to distinguish different categories. The mapped seen and unseen classes share visual and textual semantic information, which serves as anchors for knowledge transfer. Meanwhile, under the alignment constraints, visual and textual features from the same category maintain high similarity, whereas visual features are pushed away from textual features of different categories.

preliminary image encoder are flattened into a 1-D feature vector o^i. This vector is then simultaneously processed through two parallel pathways as follows.

1) It is input to a fully connected classifier layer, yielding a class confidence vector p^{cls} for seen classes. Cross-entropy loss $\mathcal{L}_{CE}$ is employed with ground-truth labels for supervised training to facilitate optimization of the preliminary image encoder.
2) The vector is projected into the spatial–text cross-modal union embedding space S^{spa}, thus, supporting subsequent cross-modal feature alignment and similarity-weighted fusion.

In the dual-branch similarity-weighted fusion mechanism, the model simultaneously extracts image and textual features, ultimately obtaining multimodal representations composed of spectral, spatial, and textual semantic features. The spectral features and textual semantic features are projected into a unified spectral–text cross-modal alignment space via projection transformations

$$\begin{aligned} o^{spe_p} &= \mathbf{W}_{spe} \cdot o^{spe} \\ o^{t_p} &= \mathbf{W}_t \cdot o^t \end{aligned} \tag{4}$$

where o^{spe_p} and o^{t_p} represent the projected spectral and text features, respectively, and $\mathbf{W}_{spe}$ and $\mathbf{W}_t$ are the corresponding projection matrices for spectral and textual features. Similarly, spatial features and text features are projected into the spatial–text cross-modal alignment space as follows:

$$\begin{aligned} o^{spa_p} &= \mathbf{W}_{spa} \cdot o^{spa} \\ o^{t_p} &= \mathbf{W}_t \cdot o^t \end{aligned} \tag{5}$$

where o^{spa_p} denotes the projected spatial features, and $\mathbf{W}_{spa}$ represents the spatial feature projection matrix. These two cross-modal spaces separately enable high-level alignment between HSI and text in the spectral and spatial dimensions. After completing cross-modal feature alignment, the spectral–text similarity matrix $\mathbf{M}^{spe}$ and spatial–text similarity matrix $\mathbf{M}^{spa}$ are computed. To further fuse spectral and spatial information, we employ a proportional weighting strategy to combine these two matrices into a fused similarity matrix $\mathbf{M}$

$$\mathbf{M} = (1 - \lambda_{sim}) \cdot sim\left(o^{spa_p}, o^{t_p}\right) + \lambda_{sim} \cdot sim\left(o^{spe_p}, o^{t_p}\right). \tag{6}$$

The resulting fused similarity matrix $\mathbf{M}$ is subsequently utilized to compute the InfoNCE [33] loss for model optimization during training. Here, λ_{sim} is a tunable hyperparameter termed the spectral similarity fusion coefficient, with a value range of $\lambda_{sim} \in [0, 1]$. sim$(\cdot)$ denotes the cosine similarity operation used to measure similarity between two vectors. In the above, o^{spa_p} and o^{spe_p} denote the spatial and spectral features after projection, respectively, and o^{t_p} denotes the projected textual features. Finally, the InfoNCE loss is combined with the cross-entropy loss through weighted summation, yielding the overall loss function

$$\mathcal{L}_{total} = \lambda_{loss} \cdot \mathcal{L}_{CE} + (1 - \lambda_{loss}) \cdot \mathcal{L}_{InfoNCE} \tag{7}$$

where λ_{loss} is a hyperparameter controlling the loss weight, termed the loss fusion coefficient, with a value range of $\lambda_{loss} \in [0, 1]$. This parameter dynamically balances supervised classification learning and contrastive learning during the training process.

D. Implementation

To clearly illustrate the training implementation process of the proposed model, Algorithm 1 presents the pseudocode for the complete training procedure of the model composed of the SPE module and SSUE module. In the SPE module, the model performs preliminary encoding of the image and separately extracts spectral and spatial features. The spectral features are encoded and projected through an independent branch, while the spatial features are utilized both for generating auxiliary classification signals and for spatial-semantic alignment.

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Algorithm 1 SpectralZero With SPE Module and SSUE Module

Input: Training datasets $D_S^P = \{I_S^P, L_S^M\}$
Output: Trained model

for all sample-pair (i_s, l_s) in D_S^P **do**
  // SSUE module
  // Extract initial visual feature
  $o^i \leftarrow \text{visualEncoder}(i_s)$
  // Extract spectral feature
  $o^{spe} \leftarrow \text{singleSpectralExtract}(o^i)$
  // Extract spatial feature and classification signal
  $(o^{spa}, p^{cls}) \leftarrow \text{spatialFeatureExtract}(o^i)$
  // Auxiliary classification signal
  $\mathcal{L}_{CE} \leftarrow \text{crossEntropyLoss}(p^{cls}, l_s)$
  $o^{spa_p} \leftarrow \text{spatialProjectionHead}(o^i)$
  $o^{spe_p} \leftarrow \text{spectralProjectionHead}(o^{spe})$

  // SPE module
  $t_s \leftarrow \text{generateEnrichmentText}(Name, Attributes)$
  $o^t \leftarrow \text{CLIPTextEncoder}(t_s)$
  $o^{t_p} \leftarrow \text{textProjectionHead}(o^t)$ // Text projection

  // Alignment (spatial–text and spectral-text)
  $M^{spa} \leftarrow \text{cosineSimilarity}(o^{t_p}, o^{spa_p})$
  $M^{spe} \leftarrow \text{cosineSimilarity}(o^{t_p}, o^{spe_p})$

  // Similarity fusion
  $M \leftarrow (1 - \lambda_{\text{sim}})\, M^{spa} + \lambda_{\text{sim}}\, M^{spe}$
  $\mathcal{L}_{\text{InfoNCE}} \leftarrow \text{InfoNCELoss}(M)$
  // Fusion loss
  $\mathcal{L}_{\text{total}} \leftarrow \lambda_{\text{loss}}\, \mathcal{L}_{CE} + (1 - \lambda_{\text{loss}})\, \mathcal{L}_{\text{InfoNCE}}$

  // Update
  $\mathcal{L}_{\text{total}}$.backward()
  update(model.parameter)
end for

The SSUE module extracts semantic features using a frozen CLIP text encoder and maps them into a unified semantic space for dual-path alignment with the image features. Based on this, two similarity matrices are constructed and fused via weighted combination, and the model is optimized for image–text alignment capability using the InfoNCE [33] loss, while the auxiliary classification signal is simultaneously fused for end-to-end training. This implementation process provides a unified training foundation for subsequent experimental settings and multimodel comparisons. The specific hyperparameter configurations will be further detailed in Section III.

III. Experiments

A. Experimental Settings

1) Datasets: We conducted experiments on three public datasets to validate the effectiveness of our model capacity in HSI classification: the Indian Pines (IP) dataset, the Houston 2013 (Houston) dataset, and the WHU-Hi-LongKou (WHHL) [34], [35] dataset. Dataset configurations are summarized in Table III, we randomly select 20% of the samples from each seen class for training, while all samples from unseen classes are used for testing. To ensure a fair and comprehensive evaluation, we adopt random rotation for the unseen classes across folds and then keep the resulting splits fixed for all experiments, such that each class is treated as unseen exactly once.

a) Indian Pines: The IP dataset was collected by the AVIRIS sensor in 1992. It consists of an HSI of size 145×145 with 224 spectral bands, covering the electromagnetic spectrum from 0.4 to 2.5 μm. Since some bands are located in water absorption regions, 24 bands (104–108, 150–163, and 220) were removed following standard practice, resulting in 200 effective bands. The image mainly covers agricultural and vegetation areas, with approximately two-thirds consisting of crop-related land covers and the remaining one-third predominantly forests or mixed vegetation. A total of 16 classes are included, such as "Woods" and "Grass Trees." We selected 13 classes for the training set, while the remaining three classes were designated as the test set. The false-color image and class labels are shown in Fig. 6.

b) Houston 2013: The Houston dataset is one of the benchmark datasets utilized in our study. It was acquired by the ITRES CASI-1500 sensor in 2013 over the University of Houston campus and its surrounding areas, comprising 144 spectral bands ranging from 0.38 to 1.05 μm. The image dimension is 349 $\times$ 1905 pixels with a spatial resolution of 2.5 m. This dataset primarily represents typical urban land-cover types, with 15 annotated classes, including "Highway" and "Commercial." We selected 12 classes as the training set, and the remaining three classes were reserved for the test set. The false-color image and class labels are illustrated in Fig. 7.

c) WHU-Hi-LongKou [34], [35]: The WHHL dataset was acquired in 2018 over Longkou Town, Hubei Province, China, using a Headwall Nano-Hyperspec hyperspectral imaging system mounted on an uncrewed aerial vehicle (UAV) platform at approximately 500-m altitude. The collected images underwent radiometric calibration and geometric correction, ensuring good physical consistency and spatial accuracy. The WHHL image consists of 270 spectral bands covering a range from 0.4 to 1.0 μm, with dimensions of 550 $\times$ 400 pixels and a spatial resolution of 0.463 m. This dataset predominantly covers agricultural areas, annotated with 9 typical land-cover classes such as "Rice" and "Cotton." We selected six classes for the training set, while the remaining three classes were used for testing. The false-color image and class labels are depicted in Fig. 8.

2) Evaluation Metrics: To comprehensively evaluate the model's performance on zero-shot classification, four standard metrics are employed: overall accuracy (OA), average accuracy (AA), Kappa coefficient, per-class accuracy, and coefficient of variation (CV). OA quantifies the model's generalization by computing the ratio of correctly predicted samples to the total number of unseen class samples. Per-class accuracy assesses class-specific recognition by measuring the proportion of correct predictions within each class, while AA reflects overall class-wise balance by averaging per-class accuracy across all categories. The Kappa coefficient further evaluates the statistical consistency between predicted and true labels,

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TABLE III

SETTINGS FOR THE IP, HOUSTON, AND WHHL DATASETS (TRAIN ON 20% OF SEEN CLASSES; THREE UNSEEN CLASSES PER FOLD FOR TESTING, NONOVERLAPPING ROTATIONS)

Group	IP			Houston			WHHL		
	Class	Total	Train(Seen)	Class	Total	Train(Seen)	Class	Total	Train(Seen)
1	CornNotill	1428	285	Healthy Grass	1251	250	Corn	34511	6902
	HayWindrowed	478	95	Synthetic Grass	697	139	Cotton	8374	1674
	Woods	1265	253	Water	325	65	Water	67056	13411
2	GrassTrees	730	146	Commercial	1244	248	Sesame	3031	606
	BuildGraTreeDri	386	77	Railway	1235	247	Broad-Leaf Soybean	63212	12642
	SoybeanMintill	2455	491	Tree	1244	248	Rice	11854	2370
3	GrassPasMow	28	15	Tennis Court	428	85	Narrow-Leaf Soybean	4151	830
	StoneSteTow	93	18	Running Track	660	132	Mixed Weed	5229	1424
	CornMintill	830	166	Parking Lot 1	1233	246	Roads And Houses	7124	1045
4	Corn	237	47	Parking Lot 2	469	93			
	Alfalfa	46	15	Residential	1268	253			
	GrassPasture	483	96	Road	1252	250			
5	Wheat	205	41	Stressed Grass	1254	250			
	Oats	20	15	Soil	1242	248			
	SoybeanNotill	972	194	Highway	1227	245			
6	SoybeanClean	593	118						
Total		10249	2072	Total	15029	2999	Total	204542	40904

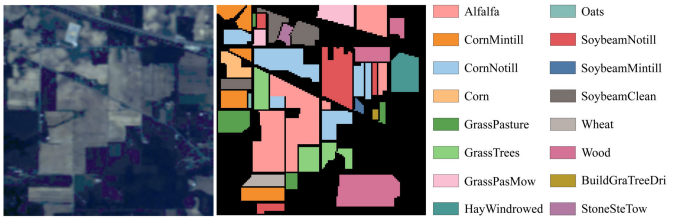


Fig. 6. False-color image and its corresponding ground-truth map of the IP dataset.

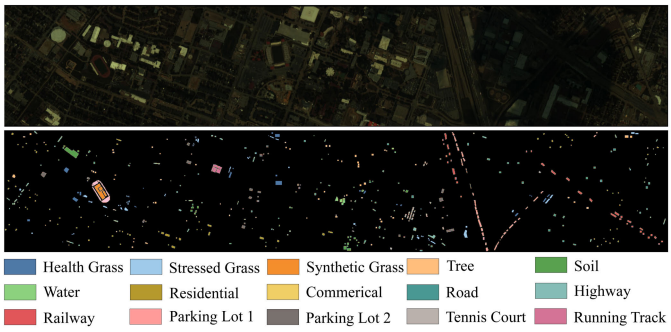


Fig. 7. False-color image and its corresponding ground-truth map of the Houston dataset.

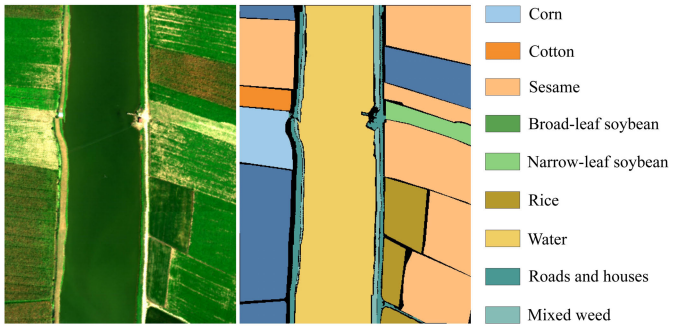


Fig. 8. False-color image and its corresponding ground-truth map of the WHHL dataset.

effectively accounting for chance agreement and enhancing the robustness of performance evaluation. To further demonstrate the stability of our model, we evaluate potential per-class bias using the CV. CV is defined as the ratio between the standard deviation and the mean of the per-class accuracies, which can be expressed as

$$\mathrm{CV} = \frac{\sigma}{\mu} \tag{8}$$

where σ and μ denote the standard deviation and mean of the per-class accuracies, respectively. A lower CV implies more balanced per-class performance, indicating better model stability.

3) Training Details: SpectralZero is implemented based on the PyTorch framework. The experiments are conducted on a machine equipped with an Intel[1] Xeon[1] Platinum 8352V CPU@2.10 GHz and a 24-GB NVIDIA GeForce RTX 4090 GPU, with 90 GB of system memory. We use GPT-4 as the LLM to generate SPE texts.

4) Parameter Setting: In this section, we describe several hyperparameter settings that have a relatively minor impact on model performance, including batch size, learning rate, number of training epochs, optimizer selection, and the temperature coefficient used in contrastive loss. To balance training efficiency with memory constraints, the batch size is uniformly set to 1024 and the train sample is set to 20% for all datasets. Based on the characteristics of each dataset, we adopt different initial learning rates: 0.0001 for IP, 0.0002 for

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TABLE IV
ZERO-SHOT CLASSIFICATION RESULTS OF EXPERIMENTS ON THE IP DATASET

Group	Unseen Classes	Model					
		f-CLSWGAN 2018	APN 2020	CE-GZSL 2021	ACMR 2022	SPECIAL 2025	Ours
Group 1	CornNotill	74.26	86.73	66.07	72.88	0.00	99.84
	HayWindrowed	30.44	14.01	65.90	0.22	0.00	94.39
	Woods	67.17	96.01	29.47	98.34	100.00	67.99
Group 2	GrassTrees	99.49	6.33	26.88	0.00	2.40	99.67
	BuildGraTreeDri	8.93	0.00	0.00	0.00	0.00	61.60
	SoybeanMintill	0.00	98.72	90.40	100.00	0.00	97.45
Group 3	GrassPasMow	100.00	69.23	0.00	0.00	0.00	100.00
	StoneSteTow	0.00	0.00	45.33	100.00	0.00	0.00
	CornMintill	0.00	98.60	64.07	0.00	0.00	97.13
Group 4	Corn	2.10	0.00	0.00	100.00	98.93	100.00
	Alfalfa	0.00	51.61	0.00	3.23	0.00	0.00
	GrassPasture	99.74	93.02	100.00	0.00	67.37	93.58
Group 5	Wheat	0.00	0.00	86.59	0.00	58.08	99.66
	Oats	100.00	92.07	0.00	100.00	0.00	100.00
	SoybeanNotill	44.22	74.68	53.98	0.00	0.00	92.08
Group 6	SoybeanClean	79.79	98.32	60.00	100.00	99.90	99.85
Results	OA (%)	42.60	74.00	57.53	30.31	29.56	**97.34**
	AA (%)	44.13	48.81	43.04	42.16	26.67	**81.45**
	$\kappa \times 100$	16.20	30.06	13.69	8.81	17.74	**87.78**

Houston, and 0.00002 for WHHL. Regarding training epochs, the maximum number of epochs is set to 100 for IP, 150 for Houston, and 50 for WHHL. The model is evaluated every 10 epochs for IP and Houston, and every 5 epochs for WHHL. Since the final performance is selected as the best among all evaluation points, minor adjustments to the number of epochs have a negligible influence on the final results. We employ the AdamW optimizer [36] across all datasets, which decouples weight decay from the gradient-based parameter updates, thereby improving model generalization. The temperature coefficient in the contrastive loss remains learnable throughout training and is initialized to 0.1 for IP and Houston and to 0.17 for WHHL. To further improve model robustness, we apply flip augmentation and random noise augmentation to all datasets during training. For patch construction, in order to enhance feature extraction quality, we set the patch size to 15, 11, and 13 for IP, Houston, and WHHL, respectively.

B. Performance Analysis

In our comparative experiments, we include not only traditional zero-shot classification models but also several representative methods specifically designed for HSI zero-shot classification. The selected baselines include: f-CLSWGAN [22], APN [37], CE-GZSL [38], ACMR [39], and SPECIAL [32].

1) f-CLSWGAN [22] as a representative approach in the field of zero-shot classification, f-CLSWGAN jointly trains a feature generator and a discriminator, incorporating an auxiliary classification loss to enhance the discriminability of the generated features. Ultimately, this method synthesizes features for unseen classes using the generator and combines them with seen class samples to train a downstream classifier, thereby enabling recognition of unseen categories.
2) APN [37] achieves zero-shot classification by aligning local image features with class attribute features, while also leveraging global image feature information for final discrimination.
3) CE-GZSL [38] proposes a hybrid generalized zero-shot classification framework that integrates feature generation with discriminative embedding. By mapping both real and synthesized samples into a contrastive embedding space and combining class-level and instance-level supervision, the model enhances classification performance. Given that the original implementation supports the ZSC setting, we train and evaluate this model under the strict zero-shot classification configuration for comparison.
4) ACMR [39] models the data distribution by constructing both visual VAEs and textual VAEs, and introduces an IEM module to prevent latent space collapse and to enhance cross-modal information integration between visual and textual modalities.
5) The SPECIAL [32] model has been detailed in Section I. As the most recently proposed method for HSI zero-shot classification, it demonstrates significant advantages in both performance and methodological design, making it a strong baseline for comparison.

To ensure a fair comparison, all methods are evaluated with the same SPE text as input; unless otherwise noted, all remaining hyperparameters and implementation details follow the best-performing settings reported in their respective papers.

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TABLE V
ZERO-SHOT CLASSIFICATION RESULTS OF EXPERIMENTS ON THE HOUSTON DATASET

Group	Unseen Classes	Model					
		f-CLSWGAN 2018	APN 2020	CE-GZSL 2021	ACMR 2022	SPECIAL 2025	Ours
Group 1	Healthy Grass	93.15	94.20	83.16	99.43	93.57	100.00
	Synthetic Grass	19.71	92.94	54.34	3.07	0.00	81.54
	Water	75.23	8.54	33.31	0.00	91.91	97.63
Group 2	Commercial	0.00	14.26	85.14	99.70	0.30	36.53
	Railway	98.99	15.89	0.00	0.00	0.00	69.60
	Tree	91.27	89.23	21.39	0.00	0.00	85.78
Group 3	Tennis Court	1.17	13.99	58.31	0.29	0.00	74.06
	Running Track	1.70	1.33	0.00	85.23	0.00	48.94
	Parking Lot 1	100.00	97.57	74.77	37.99	72.29	80.86
Group 4	Parking Lot 2	12.77	1.06	80.05	55.59	0.00	57.67
	Residential	15.07	92.41	45.62	5.32	0.00	62.06
	Road	79.24	27.05	0.00	86.73	1.90	61.54
Group 5	Stressed Grass	100.00	99.50	100.00	70.62	0.00	99.80
	Soil	0.00	98.09	0.00	0.00	0.00	0.20
	Highway	0.00	0.00	0.00	64.36	100.00	100.00
Results	OA (%)	52.09	58.64	43.78	45.05	27.49	**71.36**
	AA (%)	45.88	50.35	42.40	40.55	24.00	**70.35**
	$\kappa \times 100$	22.53	30.11	13.91	11.39	15.12	**55.79**

TABLE VI
ZERO-SHOT CLASSIFICATION RESULTS OF EXPERIMENTS ON THE WHHL DATASET

Group	Unseen Classes	Model					
		f-CLSWGAN 2018	APN 2020	CE-GZSL 2021	ACMR 2022	SPECIAL 2025	Ours
Group 1	Corn	33.93	60.04	39.57	79.13	96.60	99.26
	Cotton	37.79	63.76	59.41	3.74	49.97	85.40
	Water	98.94	97.78	98.93	98.81	99.98	100.00
Group 2	Sesame	10.14	0.00	0.00	1.79	0.00	93.70
	Broad-Leaf Soybean	87.72	100.00	1.79	23.42	0.00	1.47
	Rice	87.12	0.00	99.23	4.03	5.93	89.50
Group 3	Narrow-Leaf Soybean	74.57	0.00	77.60	0.18	1.06	95.50
	Mixed Weed	32.13	22.11	75.94	44.12	0.00	18.76
	Roads And Houses	32.22	87.84	11.46	25.23	69.08	26.00
Results	OA (%)	68.54	69.75	47.47	43.33	42.02	**76.46**
	AA (%)	54.95	47.95	51.55	31.16	35.66	**67.74**
	$\kappa \times 100$	44.74	25.15	28.14	13.53	36.75	**59.48**

Tables IV–VI present the experimental results of the proposed method and several existing zero-shot classification models on three HSI datasets: IP, Houston, and WHHL. The evaluation metrics include OA, AA, per-class classification results, and the Kappa coefficient.

As shown in Tables IV–VI, SpectralZero achieves the best zero-shot classification overall performance on all three datasets. Among the baseline methods, APN demonstrates stable performance across multiple categories, achieving particularly high OA and AA on the three datasets. On the contrary, CE-GZSL, ACMR, and SPECIAL exhibit certain limitations in classification performance for most categories. The f-CLSWGAN model performs less competitively in scenarios with fewer training samples (such as IP and Houston), which may be due to its high dependency on data scale; however, its performance improves on the WHHL dataset, where more training samples are available. Overall, the proposed method outperforms the second-best models by 23.34% (APN), 12.72% (APN), and 6.71% (APN) in OA on the IP, Houston, and WHHL datasets, respectively. For the AA, our method achieves improvements of 32.64% (APN), 20.00% (APN), and 12.79% (f-CLSWGAN) over the respective second-best models on the IP, Houston, and WHHL datasets, fully demonstrating the significant advantage of our method in zero-shot generalization capability.

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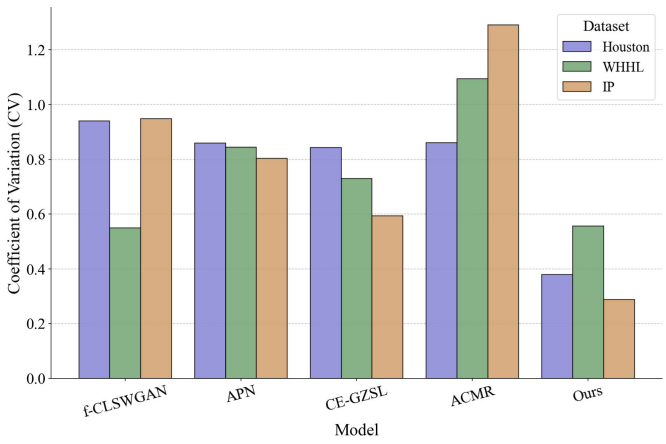


Fig. 9. CV across models on three datasets. Lower values on the y-axis indicate greater robustness.

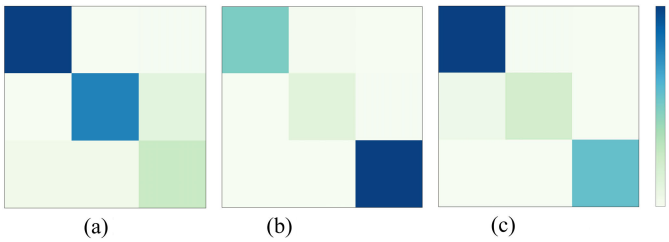


Fig. 10. Confusion matrices of the proposed model on (a) IP, (b) Houston, and (c) WHHL datasets. The rows denote the ground-truth classes and the columns denote the predicted classes. The matrices are shown in count form: diagonal entries indicate the number of correctly classified samples for each class, while off-diagonal entries indicate the number of samples misclassified into other classes, reflecting interclass confusion. Darker colors indicate higher values. This confusion matrix is computed from the first split in the three-class rotation protocol for each dataset and is provided to visually illustrate typical confusion patterns.

Furthermore, based on the per-class results in Tables IV–VI, we analyze the class-level performance across different datasets. On the IP dataset, our method achieves high recognition accuracy for most classes, e.g., 99.84% for CornNotill and 100.00% for Corn. Nevertheless, a few classes exhibit notable failures, such as 0.00% accuracy for StoneSteTow in Group 3 and 0.00% for Alfalfa in Group 4. This phenomenon is mainly attributed to the relatively limited number of effective samples for these classes (fewer than 20), as well as pronounced intraclass variations in fine-grained spectral–spatial representations for Alfalfa and Stone Steel Tower, which can destabilize class prototypes and the projected feature distribution, thereby weakening the discriminability of cross-modal alignment.

On the Houston dataset, the performance shows a more pronounced divergence in class difficulty. On the one hand, our method performs strongly on classes with relatively consistent material responses and clear spatial structures, such as healthy grass and highway, both reaching 100.00%, indicating that these classes are more likely to form stable and alignable cross-modal semantic anchors. On the other hand, under complex urban scenarios, some classes remain challenging and even suffer extreme failures—for example, commercial achieves only 36.53%, and soil drops to 0.20%. This is fundamentally due to stronger intraclass heterogeneity and more severe background mixing effects for these classes. Taking Soil as an example, it is often interwoven with bare land, construction areas, or road-edge materials, leading to mixed spectral responses and overlap with other hard-surface classes, which substantially increases the discrimination difficulty.

On the WHHL dataset, the per-class results also reveal clear class-wise disparities. Our method attains high accuracy on classes such as water and corn, while broad-leaf soybean, mixed weed, and roads and house exhibit evident failures. This behavior is closely related to their spatial distribution patterns: these difficult classes often appear as narrow strip-like regions or boundary-transition areas and are tightly adjacent to neighboring land-cover types, making patch-level representations more susceptible to surrounding background interference, which in turn induces misclassifications and exacerbates interclass confusion.

To rigorously assess class-level bias and robustness across datasets, we adopt the CV, which has been defined as in (8), as a stability metric to quantify per-class classification accuracy, the results are shown in Fig. 9. Across most settings, our method attains lower CV values on all three datasets, indicating more stable and less biased per-class performance.

In summary, the proposed model enhances the perception of spectral differences between categories by independently extracting spectral information from mixed features and constructing a semantic alignment pathway, complemented by fine-grained textual descriptions. For categories with highly similar spatial structures (e.g., “wheat” and “oats”), the ability to model spectral information enables more accurate differentiation; for categories that rely on color and spectral descriptions (e.g., “HayWindrowed”), semantic prompt texts further strengthen the model’s understanding and representation of spectral-semantic features, thereby improving overall classification performance.

To further analyze the classification capability of the proposed model for each category, Fig. 10 presents the confusion matrices of our method on the three datasets. As can be observed from the figure, most classification results are highly concentrated along the main diagonal, indicating that the model can accurately distinguish between different classes in each dataset and achieves strong overall classification performance. Meanwhile, the number of misclassifications in the off-diagonal regions of the confusion matrices is relatively small, demonstrating that the model still maintains good discriminative ability when dealing with classes that have similar spectral characteristics or spatial structures. These results validate the effectiveness and robustness of our approach in complex scenarios, and further confirm the generalization capability of the model under zero-shot settings.

C. Model Analysis

1) Ablation Study: To further evaluate the effectiveness of incorporating spectral information and semantic prompt-enhanced texts in the similarity fusion weighting, we design several ablation experiment scenarios as follows.

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TABLE VII
ABLATION STUDY RESULTS (%) ON THE IP DATASETS, HOUSTON DATASETS, AND WHHL DATASETS WITH ALL CLASSES UNDER ZERO-SHOT SETTINGS

Method	Setting	IP			Houston			WHHL		
		OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$
Baseline	Setting 1	93.35	76.90	75.79	67.63	66.99	49.78	73.60	63.24	54.08
w/o SPE module	Setting 2	94.41	80.76	80.05	68.04	65.97	49.52	74.86	66.94	55.70
w/o SSUE module	Setting 3	96.71	81.02	85.03	68.11	68.32	50.81	74.37	63.38	55.24
Full	Setting 4	**97.34**	**81.45**	**87.78**	**71.36**	**70.35**	**55.79**	**76.46**	**67.74**	**59.48**

1) *Setting 1:* Only baseline model, with no spectral–text cross-modal alignment space and no SSEB for feature extraction.
2) *Setting 2 (Ablation of SPE Module):* Replace the SPE text with the original class name. This setting is used to evaluate the contribution of rich textual prompts in representing visual features.
3) *Setting 3 (Ablation of SSUE Module):* Set similarity fusion coefficient λ_{sim} to 0. This setting is used to analyze the contribution of the spectral feature extraction capability in the SSUE module to model performance.
4) *Setting 4:* Full SpectralZero model.

The results of the ablation experiments are shown in Table VII. Based on these results, the following conclusions can be drawn.

1) Comparing the Setting 4 with the Setting 2, we observe improvements in OA of 2.93%, 3.32%, and 1.60%, and improvements in AA of 0.69%, 4.38%, and 0.80% on the IP, Houston, and WHHL datasets, respectively. These results demonstrate that semantic prompt-enhanced texts play a significant role in improving the model's ability to express the visual characteristics of land-cover types and effectively facilitate semantic alignment between image and text features.
2) Comparing the Setting 4 with the Setting 3, OA is improved by 0.63%, 3.32%, and 2.09%, AA is improved by 0.43%, 2.03%, and 4.36% on the IP, Houston, and WHHL datasets, respectively. Overall, these results validate the effectiveness of the SSUE module in constructing semantic alignment in the spectral dimension, which is particularly evident in datasets with high spectral discriminability.
3) Compared with Setting 1, Setting 4 achieves significant performance improvements on all three datasets, with OA improvements of 3.99% (IP), 3.73% (Houston), and 2.86% (WHHL), and AA improvements of 4.55% (IP), 3.36% (Houston), and 4.50% (WHHL). These results fully demonstrate that the proposed dual-branch alignment similarity weighting fusion strategy can effectively enhance the model's discriminative power and generalization performance in HSI zero-shot classification tasks.

2) Hyperparameter Analysis: This section presents an analysis of several key hyperparameters that have a critical impact on the experimental results of the model, including the similarity fusion coefficient, the number of training samples, and the loss weights.

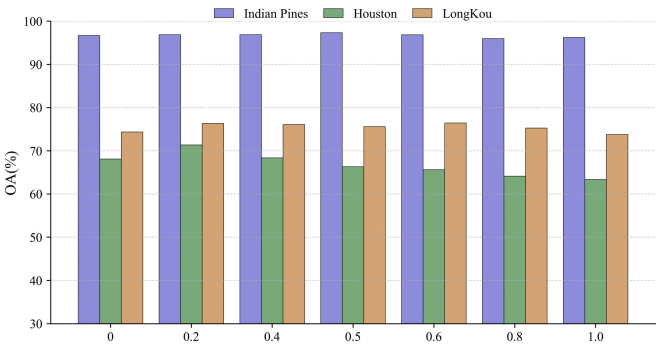


Fig. 11. Experimental results on the three datasets under different spectral similarity fusion coefficients.

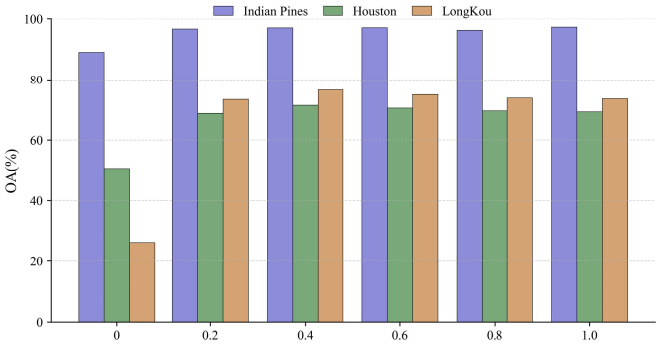


Fig. 12. Experimental results on the three datasets under different loss fusion weights.

1) *Spectral Similarity Fusion Coefficient λ_{sim}:* This coefficient is used to adjust the linear fusion ratio between the spectral–text similarity matrix S^{spe} and the spatial–text similarity matrix S^{spa}, with $\lambda_{\text{sim}} \in [0, 1]$. To evaluate the impact of this hyperparameter on model performance, we set λ_{sim} to seven values: [0, 0.2, 0.4, 0.5, 0.6, 0.8, 1] for ablation analysis, while keeping other parameters at their optimal configurations to eliminate confounding factors. As shown in Fig. 11, according to the experimental results, the model achieves optimal classification performance at λ_{sim} values of 0.2, 0.5, and 0.6 on the Houston, IP, and WHHL datasets, respectively. This result indicates that the fusion weighting of cross-modal similarity has a significant impact on the model's performance, and that the optimal value differs across datasets.

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TABLE VIII
HYPERPARAMETER SETTINGS FOR SSUE

Dataset	Image Encoder	MaxPool	SOEB			Projection Dim
			Conv3d	Conv1d	MLP	
IP	Kernel (3, 3, 3) Padding (1, 0, 0) Bias True	Kernel (1, 2, 2) Padding (0, 1, 1) Stride (4, 2, 2)	Output Channels 1 No Padding Bias True	Output Channels 4 Padding (1,) Kernel (3,)	Output Dim 128	768
Houston	Kernel (3, 1, 1) Padding (1, 0, 0) Bias True	Kernel (1, 2, 2) Padding (0, 1, 1) Stride (1, 2, 2)	Output Channels 1 No Padding Bias True	Output Channels 16 Padding (1,) Kernel (3,)	Output Dim 128	768
WHHL	Kernel (3, 3, 3) Padding (1, 0, 0) Bias False	Kernel (1, 2, 2) Padding (0, 1, 1) Stride (4, 2, 2)	Output Channels 1 No Padding Bias True	Output Channels 8 Padding (1,) Kernel (3,)	Output Dim 128	768

2) *Loss Fusion Weight λ_{loss}:* This parameter is used to adjust the linear weighting ratio between the InfoNCE loss and the auxiliary loss (cross-entropy loss) in the total loss function. By tuning the value of λ_{loss}, it is possible to balance the contributions of different loss terms and prevent one loss term from dominating the optimization process, thereby suppressing the optimization effect of other supervisory signals. To analyze the effect of this parameter on model performance, we set λ_{loss} to six values [0, 0.2, 0.4, 0.6, 0.8, 1] for analysis. The experimental results are shown in Fig. 12. According to the results, the model achieves optimal performance at λ_{loss} values of 0.4, 0.4, and 1 on the Houston, IP, and WHHL datasets, respectively. These results indicate that the coordination between different loss terms has a significant impact on both the training efficiency and final performance of the model.
3) *Other Parameters:* In addition to the main hyperparameter settings, several structural parameters are also configured in the model implementation, which are provided in Table VIII, including the kernel size for 3D convolution, the use of bias terms, and the stride settings for max-pooling operations. On the Houston dataset, the 3D convolution kernel size is set to (3, 1, 1) with corresponding padding of (1, 0, 0), and bias terms are enabled during training. For both the IP and WHHL datasets, a unified kernel size of (3, 3, 3) and padding of (1, 1, 1) are adopted. Regarding max-pooling operations, a pooling stride of (1, 2, 2) is used for the Houston dataset to preserve more spectral information, while a stride of (4, 2, 2) is used for the IP and WHHL datasets to further compress the spatial dimension and improve processing efficiency. Finally, projection dim denotes the output dimensionality of the projection heads that map the spectral and spatial branch features into the shared embedding space, which is set to 768 for all datasets.

IV. CONCLUSION

This article proposes an HSI classification method for zero-shot classification which called SpectralZero, aiming to fully exploit the semantic information in HSI and enhance the model's recognition ability for unseen classes. To achieve effective transfer of semantic knowledge, we introduce the SPE module, which extends the original class labels into natural language descriptions with visual attributes. These serve as semantic anchors for transfer, effectively bridging the semantic gap between textual and image features, and providing a more application-oriented mode of expression. To jointly model the spectral and spatial features of images, we design the SSUE module, which explicitly decouples spectral features from mixed features for independent extraction and projection, while utilizing spatial features for auxiliary classification and alignment. The two types of visual features are mapped into two independent embedding spaces and aligned with semantic features. Finally, a linear weighted fusion of similarity matrices guides the cross-modal contrastive learning process. Experimental results demonstrate that the proposed framework achieves superior classification performance compared to existing methods on three publicly available HSI datasets, verifying its effectiveness and generalizability.

In future work, we will further explore more practical settings, such as extending the model to the generalized zero-shot classification task in real-world scenarios where land-cover boundaries are ambiguous and unseen classes are unpredictable. We will also focus on improving cross-dataset generalization to better meet the transferability requirements of HSI applications.

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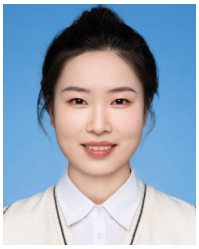
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